

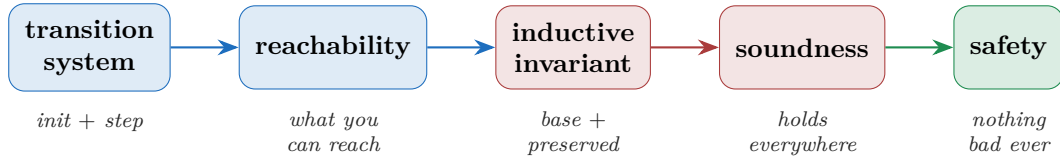
Base + Step $\Rightarrow$ Forever

The honest core of certifying model checking, in pictures

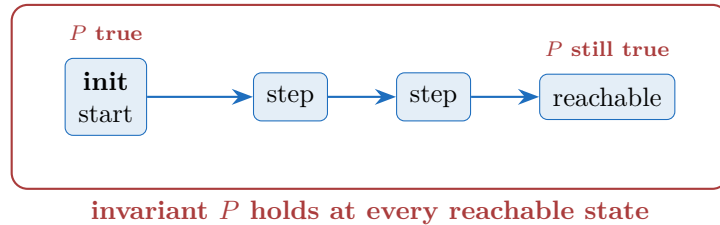
RequestProject/ModelChecking — TransitionSystem · MutexExample · TinyTS · CICLink

One idea. A property that is *true at the start* and *kept by every step* is *true forever*. That is the whole story. We (a) state it exactly, (b) let Lean check it, and (c) notice it is just Lean’s own induction rule. It is not a model checker — it is the soundness ground under one.

1. The pipeline (five words)

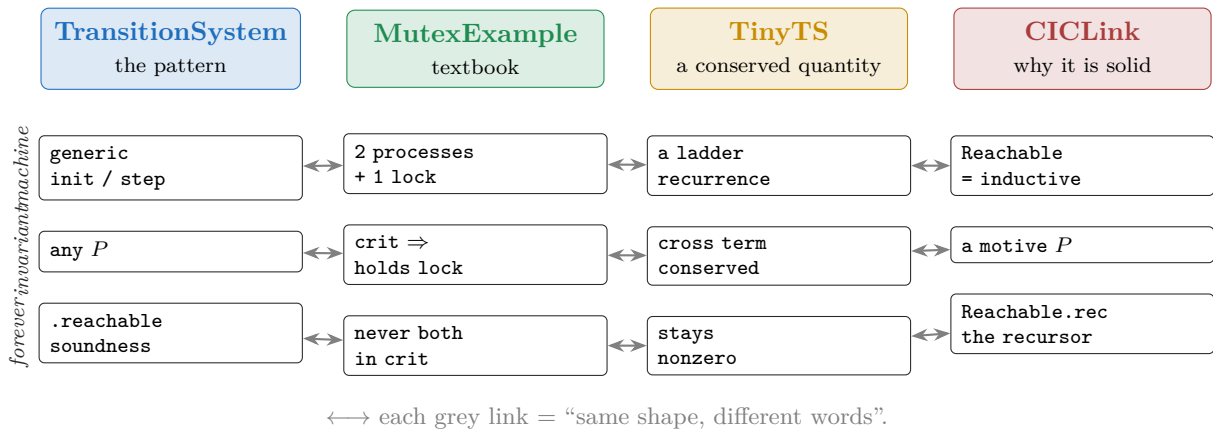


2. The picture behind it



P true at start + each step keeps $P \Rightarrow P$ true everywhere.

3. Four files, one shape

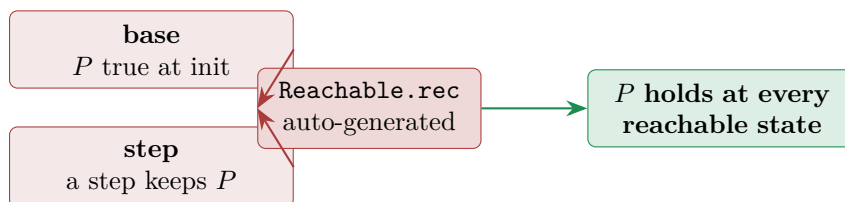


4. The two examples, side by side



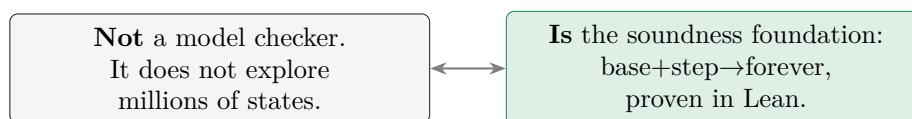
5. The elegant part: soundness *is* the recursor

```
inductive Reachable (S : System σ) : σ → Prop
| start : S.init s → Reachable S s    -- rule 1
| step : Reachable S s → S.step s s' → Reachable S s' -- rule 2
```



The two rules *are* the two constructors. Lean auto-builds the recursor `Reachable.rec`. Feed it the two invariant fields (base, step) and out comes soundness. The `by induction` proof and the raw-recursor proof are the *same term* (checked by `rfl`), and it needs *no* axioms.

6. What this is — and is not



Real tools scale this to millions of states + SMT. Here it is small on purpose — so the core is fully understood and machine-checked.

One sentence

A property true at the start and preserved by every step is true forever — stated exactly, machine-checked in Lean, and revealed to be nothing more than the inductive recursor.

All four files build with `lake build`, no `sorry`; the recursor proof uses *no* axioms.